



Lab 1-2:

Compute the reflexive transitive closure

- Implement an algorithm for computing the reflexive transitive closure R^* of any given binary relation R over some finite set $A = \{a_1, a_2, \dots, a_n\}$:
 - Input: A and R
 - Output: R^*
 - Algorithm: You can design your own algorithm or use the following one,

Initially $R^* := \emptyset$

for $i = 2, \dots, n$ do

for $i = 1, \dots, n$ do
add (a_i, a_i) to R^*

for each i -tuple $(b_1, \dots, b_i) \in A^i$ do

if $(b_1, \dots, b_i)$ is a path in R then add (b_1, b_i) to R^* .

Please briefly describe the idea of the algorithm and implement it through programming